

Assignment #6 — Milestones, Timeline, and Effort Matrix

Project: UAV Health Monitoring

Team Members:

- **Siddharth Urankar (CS):** Software, AI, Integration, Deployment
- **Prissha Chawla (CS):** Data Science, ML
- **Ally Blair (ME):** Mechanical Systems, Data Collection

1. Milestones List

#	Milestone	Description	Expected Completion
1	Motor and Propeller Data Recording Completed	Collection of baseline and faulty motor/propeller datasets under controlled lab conditions to support model training.	8 Oct
2	Initial Motor Model Developed	Development of first version of the motor health model using recorded data to detect motor faults and anomalies.	22 Oct
3	Integration on Jetson Hardware	Integrate initial motor model with onboard Jetson computer for in-flight real-time inference.	29 Oct
4	Battery Data Round 1 Collected	Perform first round of UAV battery discharge tests and record related performance metrics.	24 Oct
5	In-flight Motor Testing Conducted	Deploy and validate motor anomaly detection model in real flight-like environments.	5 Nov
6	Motor Model Fine-tuned	Improve accuracy and reduce false positives in motor fault detection through retraining.	19 Nov
7	Battery Model Prototyped	Develop an initial model for predicting battery health and remaining life using recorded data.	28 Nov
8	Strain Gauges & Second Drone Integrated	Implement strain gauges and secondary microphone array on second UAV for extended testing.	3 Dec
9	Extended Battery Data Collected	Conduct additional battery experiments to capture degradation trends under varying conditions.	31 Dec
10	Motor System Documentation Finalized	Prepare detailed documentation of test setup, model performance, and data pipeline.	31 Dec

#	Milestone	Description	Expected Completion
11	Battery Model Fine-tuned and Validated	Finalize battery model and integrate results into UAV software for full system operation.	20 Jan
12	First Design Review Completed	Present first major checkpoint on progress, design, and model performance.	1 Dec
13	Second Design Review Completed	Present full system results and progress updates before final evaluation.	20 Feb
14	Final Report and Expo Presentation Prepared	Compile final report and prepare for Expo demonstration of UAV health monitoring system.	7 Apr
15	Final Report Submission	Submit detailed project report summarizing all development, testing, and evaluation results.	24 Apr

2. Table 1 — Project Timeline

Task ID	Task Description	Start Date	End Date	Duration (days)	Associated Milestone
A	Record motor and propeller data	17-Sep	8-Oct	21	M1
B	Create initial motor model	8-Oct	22-Oct	14	M2
C	Integrate motor model with Jetson	22-Oct	29-Oct	7	M3
D	Record round 1 battery data	15-Oct	24-Oct	9	M4
E	In-flight motor testing	29-Oct	5-Nov	7	M5
F	Fine-tune motor model	7-Nov	19-Nov	12	M6
G	Decision-making on battery discharge plan	8-Oct	13-Oct	5	M4
H	Build second drone	22-Oct	5-Nov	14	M8
I	Integrate and test second microphone array	5-Nov	19-Nov	14	M8
J	Round 2 battery data collection	7-Nov	14-Nov	7	M7
K	Create initial battery model	19-Nov	28-Nov	9	M7
L	Prepare for 1st Design Review	24-Nov	1-Dec	7	M12
M	Integrate strain gauges on second drone	19-Nov	3-Dec	14	M8
N	Further battery data collection	1-Dec	31-Dec	30	M9
O	Motor documentation	1-Dec	31-Dec	30	M10
P	Test and fine-tune battery model	1-Jan	20-Jan	19	M11
Q	Prepare for 2nd Design Review	10-Feb	20-Feb	10	M13
R	Write final report	10-Feb	24-Apr	73	M15

Task ID	Task Description	Start Date	End Date	Duration (days)	Associated Milestone
S	Expo preparation and participation	24-Feb	7-Apr	42	M14

3. Table 2 — Effort Matrix

Task ID	Task Description	Ally Blair (ME)	Prissha Chawla (CS)	Siddharth Urankar (CS)	Primary Responsibility
A	Record motor and propeller data	50%	15%	35%	Ally Blair
B	Create initial motor model	10%	80%	10%	Prissha Chawla
C	Integrate motor model with Jetson	10%	10%	80%	Siddharth Urankar
D	Record round 1 battery data	50%	30%	20%	Ally Blair
E	In-flight motor testing	40%	20%	40%	Siddharth Urankar
F	Fine-tune motor model	10%	80%	10%	Prissha Chawla
G	Battery discharge planning	20%	60%	20%	Prissha Chawla
H	Build second drone	80%	10%	10%	Ally Blair
I	Integrate second microphone array	70%	10%	20%	Ally Blair
J	Record round 2 battery data	50%	30%	20%	Ally Blair
K	Create initial battery model	10%	80%	10%	Prissha Chawla
L	Prepare for 1st Design Review	33%	33%	34%	All
M	Integrate strain gauges on second drone	70%	10%	20%	Ally Blair
N	Further battery data collection	40%	40%	20%	Ally Blair
O	Motor documentation	20%	40%	40%	Siddharth Urankar
P	Fine-tune battery model	20%	70%	10%	Prissha Chawla
Q	Prepare for 2nd Design Review	33%	33%	34%	All
R	Write final report	33%	33%	34%	All

Task ID	Task Description	Ally Blair (ME)	Prissha Chawla (CS)	Siddharth Urankar (CS)	Primary Responsibility
S	Expo preparation and presentation	30%	30%	40%	Siddharth Urankar